Unit 2 - Linear Algebra
2.1 Systems of Linear Equations

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Outline

1. Why Systems of Equations?
2. Linear Equations and Their Graphs
3. The Three Outcomes
4. Solving by Hand: Substitution and Elimination
5. Three Variables
6. Gaussian Elimination
7. The Degenerate Cases
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Why Systems of Equations?

A Problem to Motivate Us

Mixing two stock solutions

"The pharmacy stocks a 10 mg/mL drug solution and a 2 mg/mL solution. We need 500 mL at 4 mg/mL. How much of each do we draw?"

Two things are unknown, so name them both. Let x be the mL of the strong solution and y the mL of the weak one. Two facts constrain them:

$$\text{the volumes must total } 500 \text{ mL:} \qquad x + y = 500$$

$$\text{the drug must total } 4 \times 500 = 2000 \text{ mg:} \qquad 10x + 2y = 2000$$

Neither equation is solvable on its own: each has two unknowns in it. Together they pin down exactly one pair. That pair, the x and y satisfying **both** equations at once, is what we mean by **solving the system**.

By the end of this lecture we will have three ways to find it, the last of which a computer can run on a system with a thousand unknowns.

Where This Shows Up in Data Science

Systems are the workhorse underneath the tools

- **Fitting a model is solving a system.** Least-squares regression reduces to a system of linear equations in the coefficients.
- **Balancing and conservation.** Kirchhoff's laws for a circuit, chemical reaction balancing, and flows through a network are all linear systems.
- **Mixtures and dosing.** Any "how much of each" question with fixed totals.
- **Redundancy detection.** A system with infinitely many solutions is telling you that your equations, or your data columns, repeat information: the same phenomenon as **collinearity** in regression.
- **Scale.** A real problem has thousands of equations. We need a procedure a machine can follow, not cleverness.

Unit 1 built **functions**. A linear equation is one of the simplest there is, and this unit asks what happens when many of them must hold at once.

What We Will Build

The plan

1. What a **linear equation** is, and why its graph is a **line**.
2. The **three outcomes**: exactly one solution, none, or infinitely many.
3. **Substitution** and **elimination**: solving by hand.
4. **Three variables**: the same ideas, with planes instead of lines.
5. **Gaussian elimination**: the **augmented matrix**, the **row operations**, **row-echelon form**, then **back-substitution**.
6. Reading **no solution** or **infinitely many** straight off the rows.
7. An **application**: ticket sales at a theater.

Notebooks for this lecture: `U2-1_Systems-1_GeometryOfSolutions`, `U2-1_Systems-2_SubstitutionAndElimination`, `U2-1_Systems-3_GaussianElimination`.

Linear Equations and Their Graphs

What Makes an Equation Linear

Definition

A **linear equation** in the variables $x_1, x_2, \ldots, x_n$ is one that can be written

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where the a_i (the **coefficients**) and b (the **right-hand side**) are fixed numbers.

Every variable appears to the **first power**, alone, multiplied only by a number. That is the whole rule.

Linear:

- $2x + 3y = 12$
- $x - y + 4z = 0$
- $7x = 21$

Not linear:

- $x^2 + y = 1$ (a power)
- $xy = 6$ (variables multiplied)
- $\sqrt{x} + y = 3$ (a root)
- $\sin(x) = y$ (inside a function)

This restriction is what makes everything in the unit work: linear problems are the ones a fixed procedure can always solve.

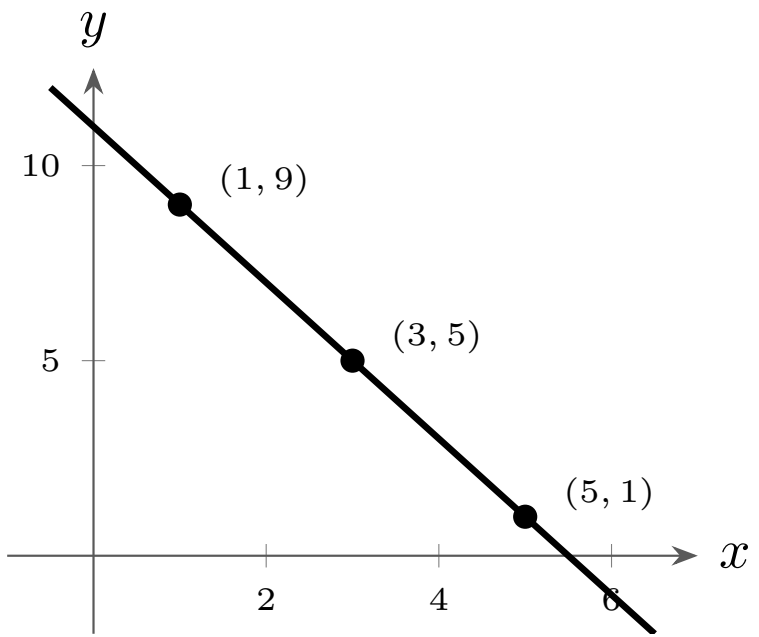
A Linear Equation in Two Variables Is a Line

Take $2x + y = 11$. Solving for y puts it in the slope-intercept form from Unit 1:

$$y \;=\; 11 - 2x$$

Every point on that line is a pair (x, y) satisfying the equation, and every pair that satisfies it lies on the line. The line **is** the solution set: not one answer, but infinitely many.

A single equation in two unknowns is therefore **underdetermined**. To pin down one pair, we need a second equation.



The line $2x + y = 11$, with three of its infinitely many solutions.

Notebook `U2-1_Systems-1_GeometryOfSolutions`, a linear equation as a line

A System Is an Intersection

Now add a second equation:

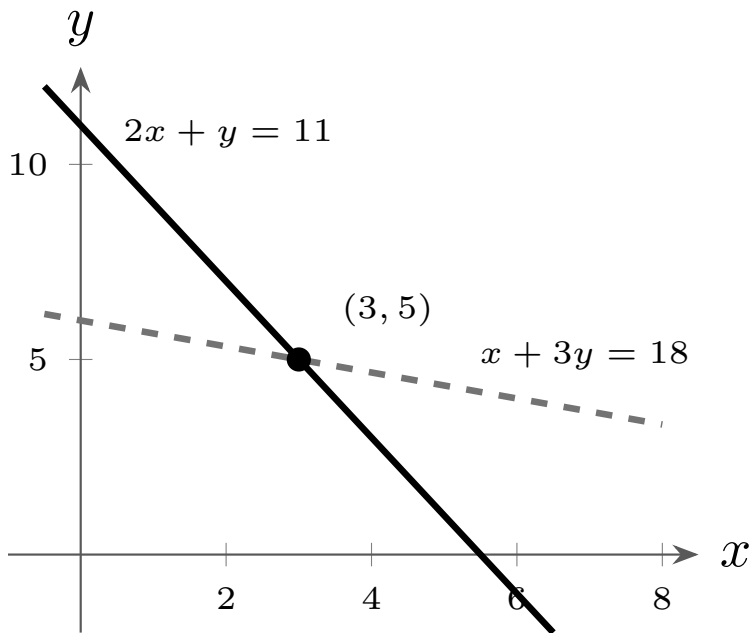
$$\begin{cases} 2x + y = 11 \\ x + 3y = 18 \end{cases}$$

A **solution of the system** satisfies **both** equations, so it must lie on **both** lines.

The geometric picture

Solving a two-variable system means finding where the lines **intersect**.

Here they meet at $(3, 5)$:
$2(3) + 5 = 11$ and $3 + 3(5) = 18$.
Always check, it costs ten seconds.

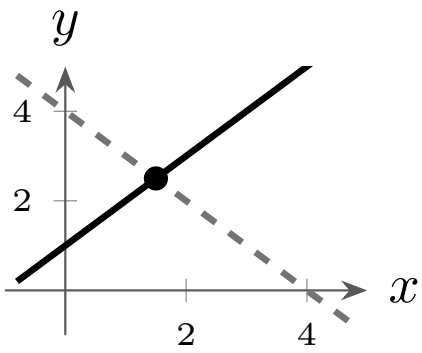


Two lines, one crossing point.

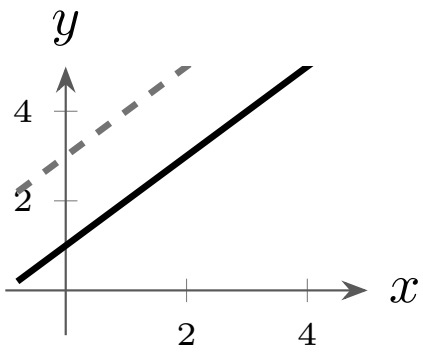
The Three Outcomes

Two Lines Can Do Only Three Things

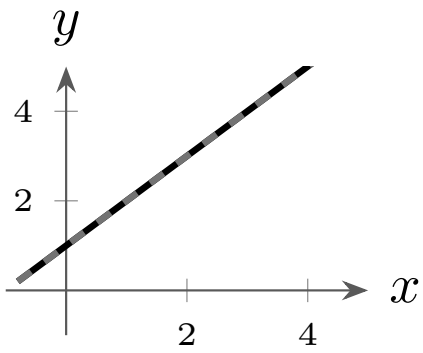
one solution



no solution



infinitely many



The trichotomy

1. **They cross once.** Exactly one solution. The system is **consistent** and **independent**.
2. **They are parallel and distinct.** No point lies on both, so there is **no solution**: the system is **inconsistent**.
3. **They are the same line.** Every point on it works, so there are **infinitely many** solutions. The second equation repeated the first: it is **dependent**.

There is no fourth case: two distinct lines cannot cross twice.

Notebook `U2-1_Systems-1_GeometryOfSolutions`, all three cases plotted

Telling Them Apart Before You Solve

Read the slopes

Put both equations in the form $y = mx + b$.

- **Different slopes**: they must cross somewhere, so exactly one solution.
- **Same slope, different intercept**: parallel, never meeting, so no solution.
- **Same slope, same intercept**: one line drawn twice, so infinitely many.

Equivalently, and this is the version that survives into three or more variables: ask whether one equation is a **multiple** of the other.

$2x + 4y = 8$ and $x + 2y = 4$:
the first is exactly 2 times the second.
Same line, infinitely many solutions.

$2x + 4y = 8$ and $x + 2y = 9$:
the left sides match up to a factor of 2,
the right sides do not.
Parallel, no solution.

Notice how little separates those two cases: only the right-hand side. Gaussian elimination will make the difference show up as one telltale row.

Exercise: Classify the System

Try it: how many solutions does each system have, and why?

1. $\begin{cases} y = 2x + 1 \\ y = 2x - 3 \end{cases}$ 2. $\begin{cases} 2x + 4y = 8 \\ x + 2y = 4 \end{cases}$ 3. $\begin{cases} x + y = 2 \\ x - y = 0 \end{cases}$

Answer from the slopes, without solving. For the one that has a single solution, go ahead and find it.

Exercise: Solution

Answers

1. **No solution.** Both slopes are 2, but the intercepts are 1 and -3: the lines are parallel and distinct, so no pair is on both. Algebraically, setting $2x + 1 = 2x - 3$ gives $1 = -3$, a contradiction.

2. **Infinitely many.** Divide the first equation by 2 and you get the second exactly, so both are the line $y = 2 - \frac{1}{2}x$. Any point on it solves the system: $(0, 2)$, $(2, 1)$, $(4, 0)$, and so on.

3. **Exactly one.** In slope form these are $y = 2 - x$ (slope -1) and $y = x$ (slope 1). Different slopes, so they cross once. Adding the two equations gives $2x = 2$, so $x = 1$ and then $y = 1$. The solution is $(1, 1)$, and it checks in both equations.

Cases 1 and 2 both have proportional left sides. Only the right-hand side decides which one you are in.

Solving by Hand

The Method of Substitution

The idea

Solve one equation for one variable, **substitute** that expression into the other, and the second equation has a single unknown.

The algorithm

1. Isolate the easiest variable (coefficient 1 is ideal).
2. Substitute into the **other** equation.
3. Solve that one-variable equation.
4. Back-substitute for the first variable.
5. Check in both originals.

Worked:

$$\begin{cases} 2x + y = 11 & (1) \\ x + 3y = 18 & (2) \end{cases}$$

$$y = 11 - 2x \qquad \text{isolate } y \text{ in (1)}$$

$$x + 3(11 - 2x) = 18 \qquad \text{substitute into (2)}$$

$$x + 33 - 6x = 18$$

$$-5x = -15$$

$$x = 3$$

$$y = 11 - 2(3) = 5 \qquad \text{back into (1)}$$

Check: $2(3) + 5 = 11$ and $3 + 3(5) = 18$. The solution is $(3, 5)$, the crossing point plotted earlier.

Notebook `U2-1_Systems-2_SubstitutionAndElimination`, this system solved both ways

The Method of Elimination

The idea

Scale an equation so a variable has **matching** coefficients in both, then subtract (or add) the equations to make it **vanish**.

Why this is allowed

Multiplying an equation by a nonzero number, or adding one equation to another, never changes the solution set: both moves are reversible, so nothing is gained or lost.

Hold on to that fact. It is the entire justification for Gaussian elimination in a moment.

Worked:

$$\begin{cases} 2x + y = 11 & (1) \\ x + 3y = 18 & (2) \end{cases}$$

Multiply (1) by 3 so both equations have $3y$, then subtract (2):

$$\begin{array}{rrl} & 6x + 3y = 33 & 3 \times (1) \\ - & x + 3y = 18 & (2) \\ \hline & 5x = 15 & \\ & x = 3 & \end{array}$$

Back into (1): $2(3) + y = 11$, so $y = 5$.

Same answer $(3, 5)$, by a different road. Elimination is the one that generalizes, so it is the method we will systematize.

Which Method, and Why It Matters

Substitution is easier when

- a variable is already isolated, or has coefficient 1;
- the system is small (two equations).

Example: $y = 3x - 1, \quad 2x + y = 9.$ The first is already solved for y, so drop it into the second:

$$\begin{aligned} 2x + (3x - 1) &= 9 \\ 5x &= 10 \;\Rightarrow\; x = 2,\; y = 5 \end{aligned}$$

Elimination is easier when

- every coefficient is awkward;
- there are three or more variables.

Example: $3x + 4y = 10, \quad 5x - 2y = 8.$ Isolating anything makes fractions; instead double the second so the y terms are opposites and add:

$$\begin{array}{rrcr} & 3x + 4y & = & 10 \\ + & 10x - 4y & = & 16 \\ \hline & 13x & = & 26 \\ & x & = 2, \; y & = 1 \end{array}$$

The real reason we prefer elimination

Substitution requires you to **look** at the system and choose well; elimination is **mechanical**, with exactly one thing to do at each step. A method with no choices in it is a method a machine can run on a thousand unknowns. Notice too that elimination never touched the letters x and y: it only scaled and added rows of **numbers**. That is the next section.

Exercise: Solve Two Ways

Try it

Solve the system

$$\begin{cases} x + 2y = 7 \\ 3x - y = 7 \end{cases}$$

1. once by **substitution**,
2. once by **elimination**,

and confirm that the two agree. Then say, without plotting, what the graph looks like.

Check your answer in **both** original equations before moving on.

Exercise: Solution

By substitution

Isolate x in (1), substitute into (2):

$$\begin{aligned} x &= 7 - 2y \\ 3(7-2y) - y &= 7 \\ 21 - 6y - y &= 7 \\ -7y &= -14 \\ y &= 2 \\ x &= 7 - 2(2) = 3 \end{aligned}$$

By elimination

Double (2) so the y terms are opposites, then add:

$$\begin{array}{rrrl} & x + 2y = & 7 & (1) \\ + & 6x - 2y = & 14 & 2 \times (2) \\ \hline & 7x & = 21 & \\ & x & = 3 & \end{array}$$

Back into (1): $3 + 2y = 7$, so $y = 2$.

Answer

Both roads give $(x, y) = (3, 2)$. Check: $3 + 2(2) = 7$ and $3(3) - 2 = 7$. The slopes are $-\frac{1}{2}$ and 3, which differ, so the graph is **two lines crossing once**, at $(3, 2)$.

Three Variables

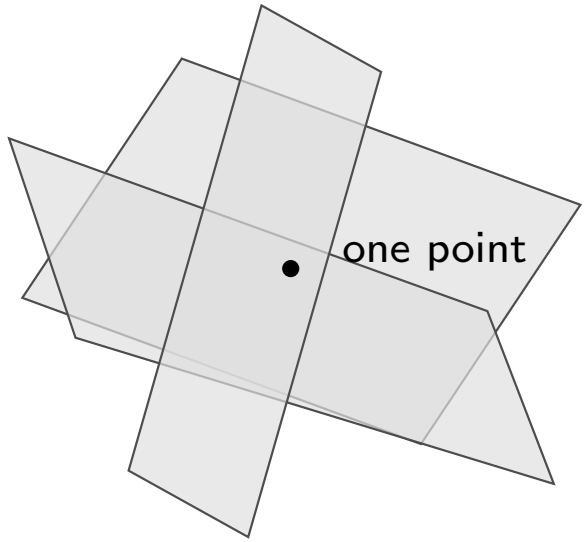
In Three Variables, an Equation Is a Plane

An equation like $2x + 3y + z = 6$ has a whole **plane** of solutions in three-dimensional space, just as a two-variable equation had a whole line.

So a system of three equations in three unknowns asks: *where do these three planes all meet?*

The outcomes are still three

Three planes can meet in a single **point** (one solution), in a whole **line** or **plane** (infinitely many), or **nowhere** at all (no solution).



Three planes crossing at a single point: the one-solution case.

Drawing this by hand gets hard fast, which is another reason to want a procedure that does not depend on picturing the geometry.

Notebook `U2-1_Systems-1_GeometryOfSolutions`, rotatable 3-D planes

Solving a Three-Variable System by Elimination

Strategy: use (1) to knock x out of (2) and (3), then finish the 2×2 and back-substitute.

$$\begin{cases} x + y + z = 6 & (1) \\ 2x - y + z = 3 & (2) \\ x + 2y - z = 2 & (3) \end{cases}$$

Eliminate x: $(2) - 2 \times (1)$ and $(3) - (1)$

$$\begin{array}{rrrrr} & 2x - & y + & z = & 3 \\ - & 2x + & 2y + & 2z = & 12 \\ \hline & -3y - & z = & -9 & (2') \\ & x + 2y - & z = & 2 & \\ - & x + & y + & z = & 6 \\ \hline & y - & 2z = & -4 & (3') \end{array}$$

Eliminate y: $(2') + 3 \times (3')$

$$\begin{array}{rrrr} -3y - & z = & -9 & \\ + \quad 3y - & 6z = & -12 & \\ \hline -7z = & -21 & & \\ z = & 3 & & \end{array}$$

Back-substitute, bottom up:

$$(3'): \quad y - 2(3) = -4 \Rightarrow y = 2$$

$$(1): \quad x + 2 + 3 = 6 \Rightarrow x = 1$$

Answer

$(x, y, z) = (1, 2, 3)$. Check all three: $1 + 2 + 3 = 6$; $2 - 2 + 3 = 3$; $1 + 4 - 3 = 2$.

The Pattern Worth Noticing

Elimination works downward, then solves upward

1. Use equation 1 to remove x from equations 2 and 3.
2. Use the **new** equation 2 to remove y from the new equation 3.
3. The newest equation 3 has only z in it: solve it for z.
4. Plug z into the new equation 2 and solve for y.
5. Plug y and z into equation 1 and solve for x.

Steps 1 and 2 are the downward sweep, **forward elimination**; steps 3 to 5 are the upward pass, **back-substitution**. The target shape is **triangular**:

$$\begin{cases} x + y + z = 6 \\ -3y - z = -9 \\ -7z = -21 \end{cases}$$

Each equation has one fewer variable than the one above it, so the last one hands you an answer for free.

Gaussian Elimination

First, Strip Away the Letters: the Augmented Matrix

The nuisance in that elimination was bookkeeping: we kept rewriting x, y, z, and $=$ when only the **numbers** changed. So let us stop writing the letters.

Definition

The **augmented matrix** of a system is the grid of its coefficients, with the right-hand sides attached as a final column, separated by a bar.

$$\begin{cases} x + y = 3 \\ 2x + y = 5 \end{cases} \quad \longleftrightarrow \quad \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 1 & 5 \end{array}\right]$$

Each **row** is one equation. Each **column** left of the bar is one variable; the column to the right holds the constants.

Watch out: a missing variable is a zero, not a blank

$$\begin{cases} x \quad + z = 4 \\ \quad y \quad = 2 \end{cases} \quad \longleftrightarrow \quad \left[\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & 0 & 2 \end{array}\right]$$

Keep the columns aligned, or the bookkeeping fails silently.

A **grid of numbers with a bar in it**, nothing more: what a matrix *is*, and what it does to a vector, waits for Unit 2-3.

Row-Echelon Form: the Target Shape

A matrix is in row-echelon form when

1. Any all-zero rows sit at the bottom.
2. In each nonzero row, the first nonzero entry (the **pivot**) is to the **right** of the pivot in the row above.
3. Every entry **below** a pivot is zero.

The result is the staircase, or triangular, shape we reached by hand. A $*$ marks an entry whose value does not matter:

$$\left[\begin{array}{ccc|c} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{array}\right] \qquad \left[\begin{array}{ccc|c} * & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{array}\right]$$

Both are valid: the pivots must move right as you go down, but need not move by exactly one column.

Gaussian elimination drives any augmented matrix into this form. The question is what we are **allowed to do** to the rows on the way: only moves that leave the solutions untouched.

The Row Operations

To reach row-echelon form we must change the numbers **without changing the answer**. These are the moves that do that:

The allowed moves, and each one is reversible

1. **Swap** two rows. $R_i \leftrightarrow R_j$
   Reordering the equations cannot change what solves them.
2. **Multiply or divide** a row by a nonzero constant. $R_i \to cR_i,\ c \neq 0$
   Doubling both sides of an equation gives an equivalent one. Not by zero, or the equation is destroyed.
3. **Add or subtract** one row to or from another. $R_i \to R_i \pm R_j$
   Adding equals to equals keeps both sides equal.
4. **Any combination** of these. e.g. $R_i \to R_i - cR_j$
   Multiply a row, then subtract it from another: the elimination step from before, and the move used most.

Systems related by these moves are **row equivalent**: they have **exactly the same solutions**. That guarantee is the whole point. We change the numbers, never the answer, only how easy it is to see.

Notebook `U2-1_Systems-3_GaussianElimination`, each operation applied step by step

The Algorithm, Stated Once

Forward elimination

1. Make **row 1, column 1** into a 1: divide the row by it (if it is 0, swap in a lower row first).
2. Make everything **below** it in that column 0: from each lower row, subtract that row's entry times row 1.
3. Move **down one row and right one column**, to the next entry on the diagonal.
4. Do the same there: make it a 1, then clear everything below it.
5. Repeat until you run out of rows. The 1's are the **pivots**.

Back-substitution

Read the rows back as equations, from the bottom up, substituting each solved variable into the row above.

No cleverness anywhere: at each step exactly one thing happens next, which is why `np.linalg.solve` can run it on systems far too large to look at.

A Worked 2×2

Solve $x + y = 3,\ \ 2x + y = 5.$

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 1 & 5 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -1 & -1 \end{array}\right] \xrightarrow{R_2 \to -R_2} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1 \end{array}\right]$$

The first row is the **pivot row**. To clear the 2 underneath the leading 1, subtract twice the pivot row from row 2. Scaling row 2 by -1 afterwards is cosmetic: it just makes the pivot a 1.

Back-substitution

Translate the rows back into equations. Row 2 reads $y = 1$. Row 1 reads $x + y = 3$, and with y known, $x = 3 - 1 = 2$.

The solution is $(x, y) = (2, 1)$. Check: $2 + 1 = 3$ and $2(2) + 1 = 5$.

Notebook `U2-1_Systems-3_GaussianElimination`, §2, every step printed with `show_aug`

A Worked 3×3: Clearing the First Column

Solve $2x + 4y + 4z = 4, \quad x + 3y + z = 4, \quad -x + 3y + 2z = -1.$

Start by scaling row 1 so the first pivot is a 1. That is optional, but it keeps the arithmetic clean.

$$\left[\begin{array}{ccc|c} 2 & 4 & 4 & 4 \\ 1 & 3 & 1 & 4 \\ -1 & 3 & 2 & -1 \end{array}\right] \xrightarrow{R_1 \to \frac{1}{2} R_1} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 1 & 3 & 1 & 4 \\ -1 & 3 & 2 & -1 \end{array}\right]$$

Now use the pivot row to knock out both entries below it.

$$\xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -1 & 2 \\ -1 & 3 & 2 & -1 \end{array}\right] \xrightarrow{R_3 \to R_3 + R_1} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -1 & 2 \\ 0 & 5 & 4 & 1 \end{array}\right]$$

The first column is done: one pivot on top, zeros beneath it. The bottom two rows are now a two-variable system in y and z, and we repeat the same move one column to the right.

A Worked 3×3: Finishing

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -1 & 2 \\ 0 & 5 & 4 & 1 \end{array}\right] \xrightarrow{R_3 \to R_3 - 5R_2} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 9 & -9 \end{array}\right]$$

That is row-echelon form: pivots at $(1,1)$, $(2,2)$, $(3,3)$, zeros below each.

Back-substitution, bottom row upward

- Row 3: $9z = -9 \;\Rightarrow\; z = -1.$
- Row 2: $y - z = 2 \;\Rightarrow\; y - (-1) = 2 \;\Rightarrow\; y = 1.$
- Row 1: $x + 2y + 2z = 2 \;\Rightarrow\; x + 2 - 2 = 2 \;\Rightarrow\; x = 2.$

The solution is $(x, y, z) = (2, 1, -1)$. Check it in the **original** three equations: $4 + 4 - 4 = 4$; $2 + 3 - 1 = 4$; $-2 + 3 - 2 = -1$. Check against the originals, not against your last matrix: that is what catches an arithmetic slip made three steps ago.

Notebook `U2-1_Systems-3_GaussianElimination`, §3, plus the `np.linalg.solve` confirmation

Exercise: Gaussian Elimination

Try it

Solve the system

$$\begin{cases} x + 2y = 5 \\ 3x + y = 10 \end{cases}$$

by writing the augmented matrix, reaching row-echelon form with row operations, and back-substituting.

Write the row operation you used above each arrow.

Then answer: which single row operation would have been needed first if the two equations had been written in the opposite order?

Exercise: Solution

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & 1 & 10 \end{array}\right] \xrightarrow{R_2 \to R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -5 & -5 \end{array}\right] \xrightarrow{R_2 \to -\frac{1}{5}R_2} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 1 \end{array}\right]$$

Back-substitution

Row 2 reads $y = 1$. Row 1 reads $x + 2y = 5$, so $x = 5 - 2(1) = 3$.

The solution is $(3, 1)$. Check: $3 + 2(1) = 5$ and $3(3) + 1 = 10$.

The follow-up. None. With the rows reversed the matrix starts as $\left[\begin{array}{cc|c} 3 & 1 & 10 \\ 1 & 2 & 5 \end{array}\right]$, whose top-left entry is 3, which is nonzero and therefore a perfectly good pivot. A **swap** is needed only when the pivot position holds a **zero**. You could still swap by choice, to get a leading 1 and easier arithmetic, but the algorithm does not require it.

The Degenerate Cases

What the Three Outcomes Look Like After Elimination

Three systems with nearly the same numbers, one row operation each. Watch the **bottom row**.

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 2 & 3 & 7 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & -1 & -1 \end{array}\right]$$

One solution. $x + 2y = 4$, $2x + 3y = 7$. Bottom row $-y = -1$: a pivot in every column, so $y = 1$, then $x = 2$.

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 2 & 4 & 9 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 0 & 1 \end{array}\right]$$

No solution. $x + 2y = 4$, $2x + 4y = 9$. Bottom row $0x + 0y = 1$, that is $0 = 1$: **false**. Stop.

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 2 & 4 & 8 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 0 & 0 \end{array}\right]$$

Infinitely many. $x + 2y = 4$, $2x + 4y = 8$. Bottom row $0 = 0$: true but **empty**. One equation left for two unknowns.

Geometrically: crossing lines, parallel lines, the same line. Elimination did not fail in the last two, it **reported**: identical left sides, and only the right-hand side decided the case.

Reading the Outcome Off the Rows

After forward elimination, look at the shape

- **A row** $\left[\begin{array}{ccc|c} 0 & \cdots & 0 & k \end{array}\right]$ **with** $k \neq 0$**:** no solution. Stop.
- **Otherwise, a pivot in every variable column:** exactly one solution. Back-substitute.
- **Otherwise, some variable column has no pivot:** infinitely many solutions, one free variable per pivot-free column.

Free variables

A column with **no pivot** is a **free variable**: set it to anything. In the third system above, y is free, so let $y = t$; row 1 then gives $x = 4 - 2t$, and every solution is

$$(x, y) = (4 - 2t,\ t) \qquad \text{for every real } t,$$

a parametric description of the line the two equations share.

The same trichotomy as the pictures of lines, but now a **mechanical check on the rows** that works in ten variables, where no picture is available.

Exercise: Read the Matrix

Try it: what does each augmented matrix say about its system?

1. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{array}\right]$

2. $\left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$

3. $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 7 \end{array}\right]$

The variables are x, y, z in that order. Where there are infinitely many solutions, describe them with a parameter.

Exercise: Solution

Answers

1. **Exactly one solution**, and it is already read off: $x = 4$, $y = -2$, $z = 3$. There is a pivot in every variable column, and each row has a single variable left in it, so there is nothing to back-substitute.

2. **Infinitely many solutions.** Columns 1 and 3 have pivots; column 2 does not, so y is **free**. Let $y = t$. Row 2 gives $z = 1$, and row 1 gives $x + 2t = 5$, so $x = 5 - 2t$. The solution set is $(5 - 2t,\ t,\ 1)$ for every real t: a line in space. The bottom row $0 = 0$ carries no information.

3. **No solution.** The bottom row says $0x + 0y + 0z = 7$, which is false for every choice of x, y, z. The rows above do not matter: an inconsistent system is inconsistent no matter what else is true.

An Application: Ticket Sales

Three Kinds of Ticket, Three Facts

A matinee

"A theater sold 300 tickets: children's at \$$6$, adults' at \$$10$, seniors' at \$$8$, and took in \$$2540$. As many adult tickets were sold as children's and seniors' combined. How many of each?"

Three unknowns, so name them: c, a, s are the numbers of children's, adults', and seniors' tickets. Each sentence of the problem is one linear equation:

$$\begin{cases} c + a + s = 300 & \text{(tickets sold)} \\ 6c + 10a + 8s = 2540 & \text{(money taken in)} \\ -c + a - s = 0 & \text{(adults = children + seniors)} \end{cases}$$

The third fact arrives as $a = c + s$; moving everything to the left puts it in the $ax + by + cz = d$ shape the augmented matrix needs, with a 0 on the right. A count, a price times count, and a comparison: three unknowns need three independent facts, and the problem supplies exactly three.

Solving It

Build the augmented matrix and clear the first column with row 1:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 6 & 10 & 8 & 2540 \\ -1 & 1 & -1 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - 6R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 4 & 2 & 740 \\ -1 & 1 & -1 & 0 \end{array}\right] \xrightarrow{R_3 \to R_3 + R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 4 & 2 & 740 \\ 0 & 2 & 0 & 300 \end{array}\right]$$

Either row could be the next pivot row. Swapping puts the smaller numbers on top and keeps everything whole:

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 2 & 0 & 300 \\ 0 & 4 & 2 & 740 \end{array}\right] \xrightarrow{R_2 \to \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 1 & 0 & 150 \\ 0 & 4 & 2 & 740 \end{array}\right] \xrightarrow{R_3 \to R_3 - 4R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 1 & 0 & 150 \\ 0 & 0 & 2 & 140 \end{array}\right]$$

Back-substitution, bottom row upward

Row 3: $2s = 140$, so $s = 70$. Row 2: $a = 150$. Row 1: $c + 150 + 70 = 300$, so $c = 80$.

Answer: 80 children's, 150 adults', 70 seniors' tickets. **Check in the originals:** $80 + 150 + 70 = 300$; $6(80) + 10(150) + 8(70) = 480 + 1500 + 560 = 2540$; $150 = 80 + 70$. All three hold.

A fractional or negative count here would not be an arithmetic slip to hide: it would say the three facts cannot all be true of a real matinee.

Back to Where We Started

The pharmacy question from the first slide

A 10 mg/mL stock and a 2 mg/mL stock; we need 500 mL at 4 mg/mL. With x mL of the strong and y mL of the weak:

$$\begin{cases} x + y = 500 & \text{(volume)} \\ 10x + 2y = 2000 & \text{(milligrams of drug)} \end{cases}$$

One elimination step and one division:

$$\left[\begin{array}{cc|c} 1 & 1 & 500 \\ 10 & 2 & 2000 \end{array}\right] \xrightarrow{R_2 \to R_2 - 10R_1} \left[\begin{array}{cc|c} 1 & 1 & 500 \\ 0 & -8 & -3000 \end{array}\right] \xrightarrow{R_2 \to -\frac{1}{8}R_2} \left[\begin{array}{cc|c} 1 & 1 & 500 \\ 0 & 1 & 375 \end{array}\right]$$

Answer

Row 2: $y = 375$. Row 1: $x = 500 - 375 = 125$. Draw 125 mL of the strong stock and 375 mL of the weak. **Check:** $125 + 375 = 500$ mL, and $10(125) + 2(375) = 2000$ mg, which is 4 mg/mL. It also makes sense: 4 is closer to 2 than to 10, so most of the mix should be the weak stock.

Notebook `U2-1_Systems-1_GeometryOfSolutions`, §3.1, the same problem solved and plotted

Review

What we covered

1. A **linear equation** allows each variable only to the first power, times a constant. In two variables its graph is a **line**; in three, a **plane**.
2. Solving a system means finding the points that satisfy **every** equation, which is where the lines or planes **intersect**.
3. There are exactly **three outcomes**: one solution (crossing), none (parallel, **inconsistent**), or infinitely many (coincident, **dependent**).
4. **Substitution** isolates a variable and feeds it into the other equation. **Elimination** scales and adds equations to make a variable vanish, and is the one that scales up.
5. The **augmented matrix** drops the letters and keeps the numbers, with the right-hand side after a bar.
6. The **row operations** (swap, multiply or divide, add or subtract, and any combination) produce **row-equivalent** systems, which have identical solutions. That is why elimination is legitimate.

Review

What we covered, continued

7. **Gaussian elimination** drives the matrix to **row-echelon form** (pivots stepping right, zeros below), then **back-substitution** reads the answer from the bottom row upward.
8. After elimination the outcome is **readable from the rows**: a row $[\,0 \;\cdots\; 0 \mid k\,]$ with $k \neq 0$ means no solution; a pivot in every variable column means one solution; a pivot-free column means a **free variable** and infinitely many.
9. The method is **mechanical**, which is exactly why `np.linalg.solve` can run it on systems far too big to draw.

Notebooks: `U2-1_Systems-1_GeometryOfSolutions`, `-2_SubstitutionAndElimination`, `-3_GaussianElimination`. **Next:** Unit 2-2, **Vectors**.